PARISHRAM 2.0 2026 (Class 12th)

CHEMISTRY

DPP 5

d & f-Block Elements

Q1 Which of the following compounds is used as the starting material for the preparation of potassium dichromate?

(A) $K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$(chrome alum)

(B) $PbCrO_4$(chrome yellow)

(C) $FeCr_2O_4$(chromite)

(D) $PbCrO_4 . PbO$(chrome red)

Q2 In the dichromate dianion,

(A) Four $Cr - O$ bonds are equivalent

(B) Six $Cr - O$ bonds are equivalent

(C) All $Cr - O$ bonds are equivalent

(D) All $Cr - O$ bonds are non-equivalent

Q3 General electronic configuration of lanthanides is

(A) $(n-2)f^{1-14}(n-1)s^2p^6d^{0-1}ns^2$

(B) $(n-2)f^{10-14}(n-1)d^{0-1}ns^2$

(C) $(n-2)f^{0-14}(n-1)d^{10}ns^2$

(D) $(n-2)d^{0-1}(n-1)f^{1-14}ns^2$

Q4 Which one of the following statements is correct when SO_2 is passed through acidified $K_2Cr_2O_7$ solution?

(A) SO_2 is reduced

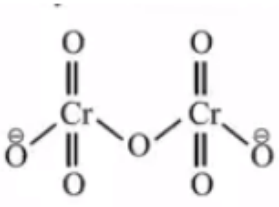
(B) Green $Cr_2(SO_4)_3$ is formed

(C) The solution turns blue

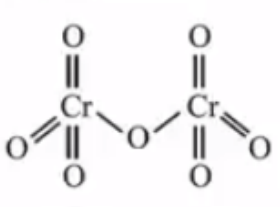
(D) The solution is decolourised

Q5 Identify the correct structure of dichromate ion:

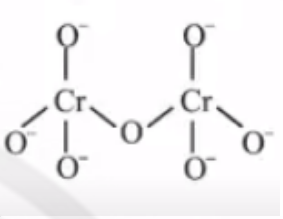
(A)



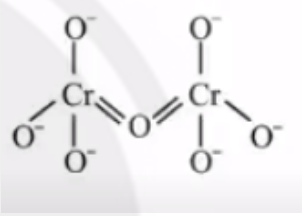
(B)



(C)



(D)



Q6 When acidified $K_2Cr_2O_7$ solution is added to Sn^{2+} salt then Sn^{2+} changes to:

(A) Sn

(B) Sn^{3+}

(C) Sn^{4+}

(D) Sn^{+}

Q7 The equilibrium $Cr_2O_7^{2-} \rightleftharpoons 2CrO_4^{2-}$ is shifted to the right in :

(A) an acidic medium

(B) a basic medium

(C) a neutral medium

(D) a weakly acidic medium

Q8 How many bridged oxygen atoms are present in chromate and dichromate anions, respectively?

(A) Zero and two (B) Zero and one

(C) One and zero (D) Two and one

Q9 The most characteristic oxidation state of lanthanides is:

(A) +2 (B) +3
(C) +4 (D) None of these

Q10 Across the lanthanide series, the basicity of the lanthanoide hydroxides
(A) Increases
(B) Decreases
(C) First increases and then decreases
(D) Does not change

Q11 The lanthanoid contraction is due to:
(A) filling of 4f orbital before the 5d orbital.
(B) introduction of lanthanoids in the second transition series.
(C) the increasing enthalpy of atomization down the groups.
(D) imperfect shielding which is caused by the 4f orbital on the electron present in 5d orbitals.

Answer Key

Q1	C
Q2	B
Q3	A
Q4	B
Q5	A
Q6	C
Q7	B
Q8	B
Q9	B
Q10	B
Q11	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Chromite ore is the industrial starting material; it is roasted with alkali and oxidised to chromate, which is subsequently processed to potassium dichromate.

Video Solution:

Q2 Text Solution:

In the dichromate ion, $Cr_2O_7^{2-}$,

- each Cr^{VI} is tetrahedrally coordinated by four O atoms;
- the two CrO_4 tetrahedra share one bridging oxygen.

Thus there are

- **six equivalent terminal Cr–O bonds** (three around each Cr), and
- **two longer, nonequivalent bridging Cr–O bonds**.

Hence **six Cr–O bonds are equivalent.**

Video Solution:

Q3 Text Solution:

General electronic configuration of lanthanides is $(n-2)d^{0-1}(n-1)f^{1-14}ns^2$

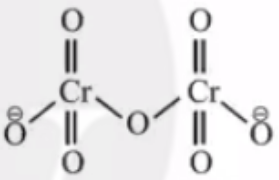
Video Solution:

Q4 Text Solution:

Green $Cr_2(SO_4)_3$ is formed

Video Solution:

Q5 Text Solution:



Video Solution:

Q6 Text Solution:

Sn^{4+}

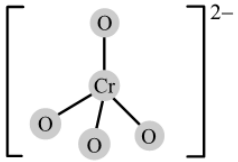
Video Solution:

Q7 Text Solution:

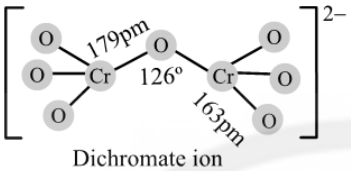
$$Cr_2O_7^{2-} \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} 2CrO_4^{2-}$$

Video Solution:

Q8 Text Solution:



Chromate ion



Dichromate ion

Video Solution:

Q9 Text Solution:

The most characteristic oxidation state of lanthanides is +3.

Video Solution:

Q10 Text Solution:

Across the lanthanide series, the basicity of the lanthanoide hydroxides decreases.

Video Solution:

Q11 Text Solution:

The lanthanoid contraction is due to imperfect shielding which is caused by the 4f orbital on the electron present in 5d orbitals.

Video Solution: